Lesson plan · for the teacher

Unplugged, then plugged

Programming · Level 4 · Lesson 1

Time
40 min

Materials
Five large cards with 1, 2, 4, 8 and 16 dots; Chalk for a sorting network (optional); One phone with this lesson downloaded

Learning goals

- Choose unplugged activities that teach real ideas.
- Link each one to code soon after.
- Judge what unplugged can and can't do.

Activity: Binary cards

1. Warm-up (5 min): Ask how many numbers five fingers can show if each finger is up or down. Collect guesses; don't answer yet.
2. Play (15 min): Five pupils hold cards with 16, 8, 4, 2 and 1 dots. Call out numbers; the class tells them which cards to turn face up. Try 13, 31 and 0.
3. Name it (5 min): Explain that each card is a bit and that face up is 1: 13 is 01101. Five bits make 32 numbers, so the answer to the warm-up is 32.
4. Plug in (10 min): Show the same numbers on one phone in the Bits and bytes lab, then the Python snippet that adds the face-up cards.
5. Wrap-up (5 min): Ask what the cards could not show (letters, bigger numbers). Then give the exit ticket.

Make the cards from cardboard, large enough to see from the back row. One phone is enough for the plugged part.

Discussion

- What did pupils understand better because they acted it out?
- What can unplugged not teach?
- How soon after should the plugged lesson come?

Exit ticket

- Name one unplugged activity and the code you will link it to.

Try it at school tomorrow
Make binary cards (1, 2, 4, 8 and 16 dots) and stand five pupils at the front. Show numbers by turning cards over; then show the same numbers on one phone in the Bits and bytes lab.

Practice questions · for you

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1 **Your class has just played binary cards. What is the best next step?**

◯ Move to a new topic ◯ Play it again next week ◯ Link it to code on one phone

2 **The halving method for binary, as Python: for 13 it prints 1101. Put the lines in order. One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐
```
bits = str(n % 2) + bits
```
☐
```
n = n % 2
```
☐
```
print(bits)
```
☐
```
n, bits = 13, ""
```
☐
```
n = n // 2
```
☐
```
while n > 0:
```

3 **The binary cards in Python. Which number do the face-up cards show?**

```
cards = [16, 8, 4, 2, 1]
up = [0, 1, 1, 0, 1]
total = 0
for i in range(5):
    total = total + cards[i] * up[i]
print(total)
```

◯ 31 ◯ 13 ◯ 11 ◯ 5

4 **Unplugged activities are enough on their own: pupils don't need to link them to code.**

◯ True ◯ False

5 **Match each unplugged activity to the computing idea it models.**

Binary cards	• • • •	Comparing and swapping to sort
A sorting network	• • • •	Exact instructions in order: an algorithm
Guess my number by halving	• • • •	How numbers are stored in bits
The human robot	• • • •	Binary search

✎ **Which unplugged activity could open your next computing unit? Write the activity, the idea it teaches, and how you will link it to code within a week.**

Check your answers

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1 **Link it to code on one phone**

Unplugged works best when it is connected to code soon after: the same idea in the Bits and bytes lab, or a few lines of Python.

2 **Top to bottom: 3, X, 5, 1, 4, 2**

```
n, bits = 13, ""
while n > 0:
    bits = str(n % 2) + bits
    n = n // 2
print(bits)
```

Each turn takes the remainder (the next bit, written in front) and then halves n.

3 **13**

The face-up cards are 8, 4 and 1, which add up to 13, or 01101 in binary.

4 **False**

Unplugged teaches the idea, but not syntax, error messages or debugging.

5 **Binary cards — How numbers are stored in bits**
A sorting network — Comparing and swapping to sort
Guess my number by halving — Binary search
The human robot — Exact instructions in order: an algorithm

Each activity is a real algorithm or data idea, acted out.